Program Check Report

Static review of saved program source against published UR guidance. No program was executed and no robot was commanded. Resolvable numeric assignments and expressions are calculated; runtime inputs, execution order and physical correctness are not inferred.

PROGRAM SUMMARY	
Program	App_DoughFactory
Software saved in URP	5.24.0
Guidance profile	CB-Series / e-Series deployment guidance
Installation source	default.installation
Installation matching	URP references ../default
Script lines / motion calls	1,920 lines; 27 movej, 34 movel, 0 movep
Other motion commands	9 optimovej, 1 optimovel
Payload / TCP source calls	8 payload; 2 TCP
Generated	2026-10-30 16:52 UTC
Overall result	17 flagged source findings; review required

SCOPE & INTERPRETATION

UR deployment acceleration values are suggested starting points for CB-Series/e-Series, not specifications or safety limits. Confirm the selected profile, physical payload/TCP, controller version and application before acting.

Only supplied source is reviewed. Saved installation settings and source-defined constants are reported separately from runtime values. Helpers, branches, URCaps and motion are not executed. Unresolved inputs remain unassessed. Installation safety serialization is not treated as an active velocity specification. Counts exclude labels/comments but include definitions; they do not indicate execution frequency.

Independent source review by UR Motion Studio. Not a Universal Robots certification or safety validation. Runtime behavior, physical setup and application suitability require verification on the applicable robot.

Review checklist

CHECKLIST	
Initial payload / CoG / inertia	Source observation
Dynamic payload changes	Manual review
Initial TCP	Source observation
Dynamic TCP changes	Manual review
Acceleration recommendations	Review recommended
Blend radius overlap	Manual review
Blended motion continuation	Manual review
Velocity specifications	Manual review
Move distances	Manual review

READ THE RESULTS

Within guideline: a resolvable classic motion acceleration is within the cited starting value. Review recommended: a concrete source value deserves attention. Source observation: a configuration call exists. Manual review: the evidence cannot establish the runtime or physical result.

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Detailed findings

Initial payload / CoG / inertia

Source observation

Saved installation payload: 3.6 kg; CoG=(0.001, 0.0, 0.132) m. URP references ../default. Saved values do not prove the physical tool setup.

UR reference [1]

Source observation · Line 3

First source occurrence; execution order and physical correctness are not proven. Mass: 3.6 kg; cog: [0.001, 0.0, 0.132] m from tool mount; inertia: [0.013017, 0.013017, 0.013017, 0.0, 0.0, 0.0] kg·m². Confirm tool plus workpiece mass, centre of gravity and inertia against the real tool; update payload when the workpiece changes.

```
set_target_payload(3.600000, [0.001000, 0.000000, 0.132000], [0.013017, 0.013017, 0.013017, 0.000000, 0.000000, 0.000000])
```

UR reference [5]

Dynamic payload changes

Source observation · Line 514

Additional source occurrence; execution order and physical correctness are not proven. Mass: 6.3 kg; cog: [0.0, -0.01, 0.1] m from tool mount; inertia: [0, 0, 0, 0, 0, 0] kg·m². Source-definition estimate using PAYLOADS (line 1261). Initialization order, runtime changes and execution are not proven. Confirm tool plus workpiece mass, centre of gravity and inertia against the real tool; update payload when the workpiece changes.

```
set_target_payload(PAYLOADS.FULL.M, PAYLOADS.FULL.COG, transition_time=0.25)
```

UR reference [5]

Independent source review by UR Motion Studio. Not a Universal Robots certification or safety validation. Runtime behavior, physical setup and application suitability require verification on the applicable robot.

Detailed findings (continued)

Source observation · Line 799

Additional source occurrence; execution order and physical correctness are not proven. Mass: 4.7 kg; cog: [0.0, -0.01, 0.1] m from tool mount; inertia: [0, 0, 0, 0, 0, 0] kg·m². Source-definition estimate using PAYLOADS (line 1261). Initialization order, runtime changes and execution are not proven. Confirm tool plus workpiece mass, centre of gravity and inertia against the real tool; update payload when the workpiece changes.

```
set_target_payload(PAYLOADS.EMPTY.M, PAYLOADS.EMPTY.COG, transition_time=0.25)
```

UR reference [5]

Source observation · Line 1535

Additional source occurrence; execution order and physical correctness are not proven. Mass: 4.7 kg; cog: [0.0, -0.01, 0.1] m from tool mount. Source-definition estimate using PAYLOADS (line 1261). Initialization order, runtime changes and execution are not proven. Confirm tool plus workpiece mass, centre of gravity and inertia against the real tool; update payload when the workpiece changes.

```
set_payload(PAYLOADS.EMPTY.M, PAYLOADS.EMPTY.COG)
```

UR reference [1]

Source observation · Line 1725

Additional source occurrence; execution order and physical correctness are not proven. Mass: 6.3 kg; cog: [0.0, -0.01, 0.1] m from tool mount. Source-definition estimate using PAYLOADS (line 1261). Initialization order, runtime changes and execution are not proven. Confirm tool plus workpiece mass, centre of gravity and inertia against the real tool; update payload when the workpiece changes.

```
set_payload(PAYLOADS.FULL.M, PAYLOADS.FULL.COG)
```

UR reference [1]

Independent source review by UR Motion Studio. Not a Universal Robots certification or safety validation. Runtime behavior, physical setup and application suitability require verification on the applicable robot.

Detailed findings (continued)

Source observation · Line 1731

Additional source occurrence; execution order and physical correctness are not proven. Mass: 4.7 kg; cog: [0.0, -0.01, 0.1] m from tool mount. Source-definition estimate using PAYLOADS (line 1261). Initialization order, runtime changes and execution are not proven. Confirm tool plus workpiece mass, centre of gravity and inertia against the real tool; update payload when the workpiece changes.

set_payload(PAYLOADS.EMPTY.M, PAYLOADS.EMPTY.COG)

UR reference [1]

Source observation · Line 1811

Additional source occurrence; execution order and physical correctness are not proven. Mass: 6.3 kg; cog: [0.0, -0.01, 0.1] m from tool mount. Source-definition estimate using PAYLOADS (line 1261). Initialization order, runtime changes and execution are not proven. Confirm tool plus workpiece mass, centre of gravity and inertia against the real tool; update payload when the workpiece changes.

set_payload(PAYLOADS.FULL.M, PAYLOADS.FULL.COG)

UR reference [1]

Source observation · Line 1817

Additional source occurrence; execution order and physical correctness are not proven. Mass: 4.7 kg; cog: [0.0, -0.01, 0.1] m from tool mount. Source-definition estimate using PAYLOADS (line 1261). Initialization order, runtime changes and execution are not proven. Confirm tool plus workpiece mass, centre of gravity and inertia against the real tool; update payload when the workpiece changes.

set_payload(PAYLOADS.EMPTY.M, PAYLOADS.EMPTY.COG)

UR reference [1]

Manual review

8 configuration call(s) in source, including definitions and branches. Call counts do not prove updates at every pickup, release or tool change. Confirm tool plus workpiece mass, centre of gravity and inertia against the real tool; update payload when the workpiece changes.

UR reference [1]

Independent source review by UR Motion Studio. Not a Universal Robots certification or safety validation. Runtime behavior, physical setup and application suitability require verification on the applicable robot.

Detailed findings (continued)

Initial TCP

Source observation

Saved installation TCP dough_factory: (0.0, 0.0, 0.247, -0.0, 3.1414, -0.0). Translation in m, rotation vector in rad. Program set_tcp calls may override this.

UR reference [1]

Source observation · Line 24

First source occurrence; execution order and physical correctness are not proven. pose: [0.0, 0.0, 0.247, -0.0, 3.1414, -0.0] (XYZ in m; rotation vector in rad). Confirm the configured TCP against the actual tool and each tool change.

```
set_tcp(p[0.0,0.0,0.247,-0.0,3.1414,-0.0])
```

UR reference [1]

Dynamic TCP changes

Source observation · Line 1536

Additional source occurrence; execution order and physical correctness are not proven. pose: [0.0, 0.0, 0.247, 0.0, 3.1414181206645937, 0.0] (XYZ in m; rotation vector in rad). Source-definition estimate using TCPS (line 1272). Initialization order, runtime changes and execution are not proven. Confirm the configured TCP against the actual tool and each tool change.

```
set_tcp(TCPS.UNDER_FLANGE)
```

UR reference [1]

Manual review

2 configuration call(s) in source, including definitions and branches. Call counts do not prove updates at every pickup, release or tool change. Confirm the configured TCP against the actual tool and each tool change.

UR reference [1]

Independent source review by UR Motion Studio. Not a Universal Robots certification or safety validation. Runtime behavior, physical setup and application suitability require verification on the applicable robot.

Detailed findings (continued)

Acceleration recommendations

Resolved accelerations are checked against entered specifications, or UR starting recommendations when the family is confirmed. Duration-based moves and runtime inputs remain unassessed. OptiMove fractions and ServoJ have distinct semantics.

Review recommended · Line 360

Configured a=3.125 m/s². Exceeds UR suggested starting value ≤2.5 m/s². This is not a safety limit or measured acceleration. Source-definition estimate using a_scale=2.5 (line 1278), al2=1.25 (line 1292). Initialization order, runtime changes and execution are not proven.

```
movel(pose_trans(pre_pick_pose, p[0.0, -0.0, PRE_PICK_GAP, 0, 0, 0]), v=vl2*v_scale, a=al2*a_scale)
```

UR reference [1]

Review recommended · Line 414

Configured a=6.25 m/s². Exceeds UR suggested starting value ≤2.5 m/s². This is not a safety limit or measured acceleration. Source-definition estimate using a_scale=2.5 (line 1278), al3=2.5 (line 1293). Initialization order, runtime changes and execution are not proven.

```
movel(pose_add(get_actual_tcp_pose(), p[0, 0, 0.015, 0, 0, 0]), v=vl3*v_scale, a=al3*a_scale)
```

UR reference [1]

Review recommended · Line 448

Configured a=3 m/s². Exceeds UR suggested starting value ≤2.5 m/s². This is not a safety limit or measured acceleration.

```
movel(retractTo, a=3.0, v=0.1)
```

UR reference [1]

Independent source review by UR Motion Studio. Not a Universal Robots certification or safety validation. Runtime behavior, physical setup and application suitability require verification on the applicable robot.

Detailed findings (continued)

Review recommended · Line 616

Configured a=3.125 m/s². Exceeds UR suggested starting value ≤2.5 m/s². This is not a safety limit or measured acceleration. Source-definition estimate using a_scale=2.5 (line 1278), al2=1.25 (line 1292). Initialization order, runtime changes and execution are not proven.

```
movel(pose_add(lift_pose, p[-DOLLY_PULL_OUT - 0.0254*12, 0.0254*8, 0.05, 0, 0, 0]), v=vl2*v_scale, a=al2*a_scale, r=0.08)
```

UR reference [1]

Review recommended · Line 623

Configured a=3.125 m/s². Exceeds UR suggested starting value ≤2.5 m/s². This is not a safety limit or measured acceleration. Source-definition estimate using a_scale=2.5 (line 1278), al2=1.25 (line 1292). Initialization order, runtime changes and execution are not proven.

```
movel(pose_add(lift_pose, p[-DOLLY_PULL_OUT, 0, 0, 0, 0, 0]), v=vl2*v_scale, a=al2*a_scale, r=0.05)
```

UR reference [1]

Review recommended · Line 637

Configured a=3.125 m/s². Exceeds UR suggested starting value ≤2.5 m/s². This is not a safety limit or measured acceleration. Source-definition estimate using a_scale=2.5 (line 1278), al2=1.25 (line 1292). Initialization order, runtime changes and execution are not proven.

```
movel(pose_add(lift_pose, p[-DOLLY_PULL_OUT - 0.0254*12, 0.0254*8, 0.05, 0, 0, 0]), v=vl2*v_scale, a=al2*a_scale, r=0.08)
```

UR reference [1]

Independent source review by UR Motion Studio. Not a Universal Robots certification or safety validation. Runtime behavior, physical setup and application suitability require verification on the applicable robot.

Detailed findings (continued)

Review recommended · Line 644

Configured a=3.125 m/s². Exceeds UR suggested starting value ≤2.5 m/s². This is not a safety limit or measured acceleration. Source-definition estimate using a_scale=2.5 (line 1278), al2=1.25 (line 1292). Initialization order, runtime changes and execution are not proven.

```
movel(pose_add(lift_pose, p[-DOLLY_PULL_OUT + 0.0254*3, 0, 0.0254*5, 0, 0, 0]), v=vl2*v_scale, a=al2*a_scale, r=0.05)
```

UR reference [1]

Review recommended · Line 737

Configured a=3.125 m/s². Exceeds UR suggested starting value ≤2.5 m/s². This is not a safety limit or measured acceleration. Source-definition estimate using a_scale=2.5 (line 1278), vl2=1.25 (line 1284). Initialization order, runtime changes and execution are not proven.

```
movel(pose_add(place_above_p, p[0.0254*0.0, -0.0254*0.25, 0, 0, 0, 0]), v=vl2*v_scale, a=vl2*a_scale, r=0.05)
```

UR reference [1]

Review recommended · Line 741

Configured a=3.125 m/s². Exceeds UR suggested starting value ≤2.5 m/s². This is not a safety limit or measured acceleration. Source-definition estimate using a_scale=2.5 (line 1278), vl2=1.25 (line 1284). Initialization order, runtime changes and execution are not proven.

```
movel(pose_add(place_above_p, p[0.0254*0.0, -0.0254*2, 0, 0, 0, 0]), v=vl2*v_scale, a=vl2*a_scale, r=0.05)
```

UR reference [1]

Independent source review by UR Motion Studio. Not a Universal Robots certification or safety validation. Runtime behavior, physical setup and application suitability require verification on the applicable robot.

Detailed findings (continued)

Review recommended · Line 744

Configured a=3.125 m/s². Exceeds UR suggested starting value ≤2.5 m/s². This is not a safety limit or measured acceleration. Source-definition estimate using a_scale=2.5 (line 1278), al2=1.25 (line 1292). Initialization order, runtime changes and execution are not proven.

movel(modified_place_q, v=vl2*v_scale, a=al2*a_scale, r=0)

UR reference [1]

Review recommended · Line 797

Configured a=4.36332 m/s². Exceeds UR suggested starting value ≤2.5 m/s². This is not a safety limit or measured acceleration. Source-definition estimate using a_scale=2.5 (line 1278), ar2=1.74532925 (line 1296). Initialization order, runtime changes and execution are not proven.

movel(pose_add(current_pose, p[0, place_y, -0.0, 0, 0, 0]), v=vr2*v_scale, a=ar2*a_scale, r=0.00)

UR reference [1]

Review recommended · Line 807

Configured a=4.36332 m/s². Exceeds UR suggested starting value ≤2.5 m/s². This is not a safety limit or measured acceleration. Source-definition estimate using a_scale=2.5 (line 1278), ar2=1.74532925 (line 1296). Initialization order, runtime changes and execution are not proven.

movel(pose_add(current_pose, p[0, place_y, -0.005, 0, 0, 0]), v=vr2*v_scale, a=ar2*a_scale, r=0.0025)

UR reference [1]

Review recommended · Line 826

Configured a=4.36332 m/s². Exceeds UR suggested starting value ≤2.5 m/s². This is not a safety limit or measured acceleration. Source-definition estimate using a_scale=2.5 (line 1278), ar2=1.74532925 (line 1296). Initialization order, runtime changes and execution are not proven.

movel(transition_pose, v=vr2*v_scale, a=ar2*a_scale, r=0.0025)

UR reference [1]

Independent source review by UR Motion Studio. Not a Universal Robots certification or safety validation. Runtime behavior, physical setup and application suitability require verification on the applicable robot.

Detailed findings (continued)

Review recommended · Line 828

Configured a=4.36332 m/s². Exceeds UR suggested starting value ≤2.5 m/s². This is not a safety limit or measured acceleration. Source-definition estimate using a_scale=2.5 (line 1278), ar2=1.74532925 (line 1296). Initialization order, runtime changes and execution are not proven.

movel(pose_trans(transition_pose, p[0,-0.021,0.042,0,0,0]), v=vr2*v_scale, a=ar2*a_scale, r=0.0025)

UR reference [1]

Review recommended · Line 843

Configured a=2.61799 m/s². Exceeds UR suggested starting value ≤2.5 m/s². This is not a safety limit or measured acceleration. Source-definition estimate using a_scale=2.5 (line 1278), ar1=1.04719755 (line 1295). Initialization order, runtime changes and execution are not proven.

movel(pre_idle_p, v=vr1*v_scale, a=ar1*a_scale, r=0.01)

UR reference [1]

Review recommended · Line 1747

Configured a=3.125 m/s². Exceeds UR suggested starting value ≤2.5 m/s². This is not a safety limit or measured acceleration. Source-definition estimate using a_scale=2.5 (line 1278), al2=1.25 (line 1292). Initialization order, runtime changes and execution are not proven.

movel(pose_add(pull_out_pose, p[0.02, 0, 0, 0, 0, 0]), v=v_scale*vl2, a=a_scale*al2)

UR reference [1]

Review recommended · Line 1835

Configured a=2.61799 m/s². Exceeds UR suggested starting value ≤2.5 m/s². This is not a safety limit or measured acceleration. Source-definition estimate using a_scale=2.5 (line 1278), ar1=1.04719755 (line 1295). Initialization order, runtime changes and execution are not proven.

movel(pre_idle_p, v=vr1*v_scale, a=ar1*a_scale, r=0.0)

UR reference [1]

Independent source review by UR Motion Studio. Not a Universal Robots certification or safety validation. Runtime behavior, physical setup and application suitability require verification on the applicable robot.

Detailed findings (continued)

SOURCE PARAMETER EVIDENCE · ACCELERATION RECOMMENDATIONS	
Lines 138, 618, 657, 664, 672, 679	Manual review · Configured a=6.10865 rad/s². Within UR suggested starting value ≤13.9626 rad/s². This is not a safety limit or measured acceleration. Source-definition estimate using a_scale=2.5 (line 1278), ar3=2.44346095 (line 1297). Initialization order, runtime changes and execution are not proven.
Lines 156, 158	Manual review · Configured a=0.8 fraction. Within the documented OptiMove fraction range (0, 1]. Absolute velocity/acceleration varies with the trajectory; physical limits are not established.
Lines 161, 307	Manual review · Configured a=0.6 fraction. Within the documented OptiMove fraction range (0, 1]. Absolute velocity/acceleration varies with the trajectory; physical limits are not established.
Lines 167	Within guideline · Configured a=1 m/s². Within UR suggested starting value ≤2.5 m/s². This is not a safety limit or measured acceleration.
Lines 199	Within guideline · Configured a=0.05 m/s². Within UR suggested starting value ≤2.5 m/s². This is not a safety limit or measured acceleration.
Lines 294, 296, 303, 639	Manual review · Configured a=1 fraction. Within the documented OptiMove fraction range (0, 1]. Absolute velocity/acceleration varies with the trajectory; physical limits are not established.
Lines 327, 577, 591, 599, 707, 1758, 1765, 1782, 1787, 1791, 1793	Manual review · Configured a=1.25 m/s². Within UR suggested starting value ≤2.5 m/s². This is not a safety limit or measured acceleration. Source-definition estimate using a_scale=2.5 (line 1278), al1=0.5 (line 1291). Initialization order, runtime changes and execution are not proven.
Lines 329	Manual review · Configured a=2.61799 rad/s². Within UR suggested starting value ≤13.9626 rad/s². This is not a safety limit or measured acceleration. Source-definition estimate using a_scale=2.5 (line 1278), ar1=1.04719755 (line 1295). Initialization order, runtime changes and execution are not proven.
Lines 390	Within guideline · Configured a=0.2 m/s². Within UR suggested starting value ≤2.5 m/s². This is not a safety limit or measured acceleration.
Lines 427	Within guideline · Configured a=1.2 m/s². Within UR suggested starting value ≤2.5 m/s². This is not a safety limit or measured acceleration.
Lines 482	Within guideline · Configured a=0.1 m/s². Within UR suggested starting value ≤2.5 m/s². This is not a safety limit or measured acceleration.

Independent source review by UR Motion Studio. Not a Universal Robots certification or safety validation. Runtime behavior, physical setup and application suitability require verification on the applicable robot.

SOURCE PARAMETER EVIDENCE · ACCELERATION RECOMMENDATIONS (continued)	
Lines 711	Manual review · Configured a=4.36332 rad/s². Within UR suggested starting value ≤13.9626 rad/s². This is not a safety limit or measured acceleration. Source-definition estimate using a_scale=2.5 (line 1278), ar2=1.74532925 (line 1296). Initialization order, runtime changes and execution are not proven.
Lines 761	Within guideline · Configured a=0.3 m/s². Within UR suggested starting value ≤2.5 m/s². This is not a safety limit or measured acceleration.
Lines 831	Manual review · Configured a=4.36332 rad/s². Within UR suggested starting value ≤13.9626 rad/s². This is not a safety limit or measured acceleration. Source-definition estimate using a_scale=2.5 (line 1278), vr2=1.74532925 (line 1288). Initialization order, runtime changes and execution are not proven.
Lines 858, 1806	Manual review · Configured a=0.5 fraction. Within the documented OptiMove fraction range (0, 1]. Absolute velocity/acceleration varies with the trajectory; physical limits are not established.
Lines 865, 1471, 1473, 1476, 1478, 1481, 1483, 1485, 1487, 1490, 1492, 1494, 1496, 1498, 1500, 1502	Within guideline · Configured a=1.39626 rad/s². Within UR suggested starting value ≤13.9626 rad/s². This is not a safety limit or measured acceleration.
Lines 1801, 1803	Manual review · Configured a=2.44346 rad/s². Within UR suggested starting value ≤13.9626 rad/s². This is not a safety limit or measured acceleration. Source-definition estimate using ar3=2.44346095 (line 1297). Initialization order, runtime changes and execution are not proven.

Blend radius overlap

Resolved blend radii are compared with known adjacent move distances and radius sums. Geometric warnings do not prove skipped waypoints; motion version and actual execution still matter.

Manual review
line 22: motion_version_set(2) Version 2 can reduce blend radii during planning. A static geometric overlap must not be presented as a proven skipped waypoint. UR reference [3]

SOURCE PARAMETER EVIDENCE · BLEND RADIUS OVERLAP	
Lines 138, 831, 1782, 1787, 1791	Manual review · Configured radius r=0.02 m. Source geometry does not prove a skipped waypoint; controller planning and execution require verification.

Independent source review by UR Motion Studio. Not a Universal Robots certification or safety validation. Runtime behavior, physical setup and application suitability require verification on the applicable robot.

SOURCE PARAMETER EVIDENCE · BLEND RADIUS OVERLAP (continued)	
Lines 156, 618, 1801, 1803	Manual review · Configured radius r=0.1 m. Source geometry does not prove a skipped waypoint; controller planning and execution require verification.
Lines 158, 294, 639	Manual review · Configured radius r=0.15 m. Source geometry does not prove a skipped waypoint; controller planning and execution require verification.
Lines 161, 167, 199, 307, 327, 360, 390, 414, 427, 448, 482, 577, 591, 707, 711, 744, 761, 797, 865, 1471, 1473, 1476, 1478, 1481	Source observation · Configured radius r=0 m. Source geometry does not prove a skipped waypoint; controller planning and execution require verification.
Lines 1483, 1485, 1487, 1490, 1494, 1496, 1498, 1500, 1502, 1747, 1806, 1835	Source observation · Configured radius r=0 m. Source geometry does not prove a skipped waypoint; controller planning and execution require verification.
Lines 296	Manual review · Configured radius r=0.25 m. Source geometry does not prove a skipped waypoint; controller planning and execution require verification.
Lines 303	Manual review · Configured radius r=0.12 m. Source geometry does not prove a skipped waypoint; controller planning and execution require verification.
Lines 329, 843, 1793	Manual review · Configured radius r=0.01 m. Source geometry does not prove a skipped waypoint; controller planning and execution require verification.
Lines 599	Manual review · Configured radius r=0.003 m. Source geometry does not prove a skipped waypoint; controller planning and execution require verification.
Lines 616, 637	Manual review · Configured radius r=0.08 m. Source geometry does not prove a skipped waypoint; controller planning and execution require verification.
Lines 623, 644, 657, 664, 672, 679, 737, 741, 858, 1492, 1758, 1765	Manual review · Configured radius r=0.05 m. Source geometry does not prove a skipped waypoint; controller planning and execution require verification.
Lines 807	Manual review · Configured radius r=0.0025 m. Source geometry does not prove a skipped waypoint; controller planning and execution require verification.
Lines 826	Manual review · Configured radius r=0.0025 m. Outgoing move distance=0.0469574 m. Source geometry does not prove a skipped waypoint; controller planning and execution require verification.

Independent source review by UR Motion Studio. Not a Universal Robots certification or safety validation. Runtime behavior, physical setup and application suitability require verification on the applicable robot.

SOURCE PARAMETER EVIDENCE · BLEND RADIUS OVERLAP (continued)	
Lines 828	Manual review · Configured radius r=0.0025 m. Incoming move distance=0.0469574 m. Adjacent radii total=0.005 m. Source geometry does not prove a skipped waypoint; controller planning and execution require verification.

Blended motion continuation

Manual review

35 motion calls have nonzero or unresolved radii. Verify continuation through branches, waits, stops and helper functions on the applicable controller. Static source inspection does not establish the executed path.

UR reference [4]

Velocity specifications

Resolved velocities are checked against entered specifications. No generic robot velocity limit is assumed. Source values do not establish measured speed.

Manual review

Joint and tool speed limits depend on the configured Normal/Reduced safety modes and application. Saved installation safety serialization is not interpreted as an active speed specification. Advanced overrides can supply verified limits.

UR reference [10]

SOURCE PARAMETER EVIDENCE · VELOCITY SPECIFICATIONS	
Lines 138, 618, 657, 664, 672, 679, 1801, 1803	Manual review · Configured v=9.77384 rad/s. Enter the applicable joint velocity specification to check this value. Source-definition estimate using v_scale=4 (line 1280), vr3=2.44346095 (line 1289). Initialization order, runtime changes and execution are not proven.
Lines 156, 158	Manual review · Configured v=0.85 fraction. Within the documented OptiMove fraction range (0, 1]. Absolute velocity/acceleration varies with the trajectory; physical limits are not established.
Lines 161	Manual review · Configured v=0.7 fraction. Within the documented OptiMove fraction range (0, 1]. Absolute velocity/acceleration varies with the trajectory; physical limits are not established.
Lines 167	Manual review · Configured v=1 m/s. Enter the applicable TCP velocity specification to check this value.

Independent source review by UR Motion Studio. Not a Universal Robots certification or safety validation. Runtime behavior, physical setup and application suitability require verification on the applicable robot.

SOURCE PARAMETER EVIDENCE · VELOCITY SPECIFICATIONS (continued)	
Lines 199	Manual review · Configured v=0.05 m/s. Enter the applicable TCP velocity specification to check this value.
Lines 294, 296, 303, 639	Manual review · Configured v=1 fraction. Within the documented OptiMove fraction range (0, 1]. Absolute velocity/acceleration varies with the trajectory; physical limits are not established.
Lines 307, 1806	Manual review · Configured v=0.5 fraction. Within the documented OptiMove fraction range (0, 1]. Absolute velocity/acceleration varies with the trajectory; physical limits are not established.
Lines 327, 577, 591, 599, 707, 1758, 1765, 1782, 1787, 1791, 1793	Manual review · Configured v=2 m/s. Enter the applicable TCP velocity specification to check this value. Source-definition estimate using v_scale=4 (line 1280), vl1=0.5 (line 1283). Initialization order, runtime changes and execution are not proven.
Lines 329	Manual review · Configured v=4.18879 rad/s. Enter the applicable joint velocity specification to check this value. Source-definition estimate using v_scale=4 (line 1280), vr1=1.04719755 (line 1287). Initialization order, runtime changes and execution are not proven.
Lines 360, 616, 623, 637, 644, 737, 741, 744, 1747	Manual review · Configured v=5 m/s. Enter the applicable TCP velocity specification to check this value. Source-definition estimate using v_scale=4 (line 1280), vl2=1.25 (line 1284). Initialization order, runtime changes and execution are not proven.
Lines 390	Manual review · Configured v=0.2 m/s. Enter the applicable TCP velocity specification to check this value.
Lines 414	Manual review · Configured v=10 m/s. Enter the applicable TCP velocity specification to check this value. Source-definition estimate using v_scale=4 (line 1280), vl3=2.5 (line 1285). Initialization order, runtime changes and execution are not proven.
Lines 427	Manual review · Configured v=0.25 m/s. Enter the applicable TCP velocity specification to check this value.
Lines 448	Manual review · Configured v=0.1 m/s. Enter the applicable TCP velocity specification to check this value.
Lines 482	Manual review · Configured v=1.5 m/s. Enter the applicable TCP velocity specification to check this value.
Lines 711, 831	Manual review · Configured v=6.98132 rad/s. Enter the applicable joint velocity specification to check this value. Source-definition estimate using v_scale=4 (line 1280), vr2=1.74532925 (line 1288). Initialization order, runtime changes and execution are not proven.

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SOURCE PARAMETER EVIDENCE · VELOCITY SPECIFICATIONS (continued)	
Lines 761	Manual review · Configured v=0.3 m/s. Enter the applicable TCP velocity specification to check this value.
Lines 797, 807, 826, 828	Manual review · Configured v=6.98132 m/s. Enter the applicable TCP velocity specification to check this value. Source-definition estimate using v_scale=4 (line 1280), vr2=1.74532925 (line 1288). Initialization order, runtime changes and execution are not proven.
Lines 843, 1835	Manual review · Configured v=4.18879 m/s. Enter the applicable TCP velocity specification to check this value. Source-definition estimate using v_scale=4 (line 1280), vr1=1.04719755 (line 1287). Initialization order, runtime changes and execution are not proven.
Lines 858	Manual review · Configured v=0.4 fraction. Within the documented OptiMove fraction range (0, 1]. Absolute velocity/acceleration varies with the trajectory; physical limits are not established.
Lines 865, 1471, 1473, 1476, 1478, 1481, 1483, 1485, 1487, 1490, 1492, 1494, 1496, 1498, 1500, 1502	Manual review · Configured v=1.0472 rad/s. Enter the applicable joint velocity specification to check this value.

Move distances

Distances are unblended translational lengths for resolvable adjacent source targets. Circular moves use arc length. Shared rigid-frame offsets can establish separation. Unknown starts, joint-space paths and uncertain runtime flow remain unassessed.

Source observation
Known source path distance=0.0469574 m across 1 of 71 move(s). This sums resolvable unblended translational segments only; unknown segments, first approach, rotations, branches and repetitions are excluded. It is not total executed travel. UR reference [4]

SOURCE PARAMETER EVIDENCE · MOVE DISTANCES	
Lines 138, 156, 158, 161, 167, 199, 294, 296, 303, 307, 327, 329, 360, 390, 414, 427, 448, 482, 577, 591, 599, 616, 618, 623	Manual review · Move distance unresolved: initial pose, joint-space trajectory, target/frame expression or intervening control flow is unknown.

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SOURCE PARAMETER EVIDENCE · MOVE DISTANCES (continued)	
Lines 637, 639, 644, 657, 664, 672, 679, 707, 711, 737, 741, 744, 761, 797, 807, 826, 831, 843, 858, 865, 1471, 1473, 1476, 1478	Manual review · Move distance unresolved: initial pose, joint-space trajectory, target/frame expression or intervening control flow is unknown.
Lines 1481, 1483, 1485, 1487, 1490, 1492, 1494, 1496, 1498, 1500, 1502, 1747, 1758, 1765, 1782, 1787, 1791, 1793, 1801, 1803, 1806, 1835	Manual review · Move distance unresolved: initial pose, joint-space trajectory, target/frame expression or intervening control flow is unknown.
Lines 828	Source observation · Unblended move distance=0.0469574 m from preceding source waypoint (translation only). Derived from a rigid offset of the same named pose; runtime pose stability and execution require verification. Source-definition substitution is conditional on initialization and unchanged values.

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Sources & next steps

Confirm matching URP/script versions. Review flagged lines in PolyScope, verify physical payload and TCP, and validate motion using the relevant robot/software manuals. Use flight measurements and incident logs to investigate observed behavior separately.

INPUT SOURCE

/home/apex/workspace/motionSim/tests/full-backup.tar ::
/programs/App_DoughFactory/App_DoughFactory.urp

SHA-256: 81e303f2b98254bd259a53cf03e0961e112d9ee66d7a67f3e2194b925c50133b

INPUT SOURCE

/home/apex/workspace/motionSim/tests/full-backup.tar ::
/programs/App_DoughFactory/App_DoughFactory.script

SHA-256: 7366e9a00551a3459df57b787d1cfdb21ce15b62fc0d02926533fbd25f89cd5e

INPUT SOURCE

/home/apex/workspace/motionSim/tests/full-backup.tar :: /programs/default.installation

SHA-256: 37475581ef1f1a022997dbc9ba9622de43e4da4355a91d7c00b5dfd90879869e

[1] UR Important Deployment Points (CB-Series/e-Series)
https://www.universal-robots.com/articles/ur/robot-care-maintenance/important-deployment-points/

[2] UR PolyScope 5.24 Script Manual: MoveJ
https://www.universal-robots.com/manuals/EN/HTML/SW5_24/Content/prod-scriptmanual/all_scripts/movej_qa14v105t0r.htm

[3] UR Motion Version (PolyScope 5.22+)
https://www.universal-robots.com/manuals/EN/HTML/SW5_22/Content/prod-scriptmanual/all_scripts/motion_version_desc.htm

[4] UR Blend Parameters
https://www.universal-robots.com/manuals/EN/HTML/SW5_19/Content/prod-usr-man/software/PolyScope/content/BasicProgNodes/commandtab_way_blend_para.htm

[5] UR Payload configuration (PolyScope 5.24)
https://www.universal-robots.com/manuals/EN/HTML/SW5_24/Content/prod-scriptmanual/all_scripts/set_target_payload.htm

[6] UR OptiMove parameter ranges
https://www.universal-robots.com/manuals/EN/HTML/SW5_24/Content/prod-scriptmanual/all_scripts/movej_optimove.htm

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UR references (continued)

[7] UR MoveC geometry and units

https://www.universal-robots.com/manuals/EN/HTML/SW5_24/Content/prod-scriptmanual/all_scripts/movec_pose_via_pose_toa1.htm

[8] UR SpeedL velocity and acceleration

https://www.universal-robots.com/manuals/EN/HTML/SW5_24/Content/prod-scriptmanual/all_scripts/speedl_xd_a_t_aRot_a.htm

[9] UR ServoJ parameters

https://www.universal-robots.com/manuals/EN/HTML/SW5_24/Content/prod-scriptmanual/all_scripts/servoj_qavt0-008lookahead_time.htm

[10] UR configurable joint and tool speed limits

https://www.universal-robots.com/manuals/EN/HTML/SW5_25/Content/prod-usr-man/complianceUR7e/H_g5_sections/safetyFunctionsAndinterfaces/configurable_functions.htm

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